

PORT NOLLOTH, South Africa

WMO: 684080

Lat: 29.2500S

Lon: 16.8686E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	7.0	-4.8	2.5	21.0	-2.1	3.2	20.3	8.4	14.7	7.9	14.6	2.3	100	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	28.5	15.3	25.5	15.1	23.4	15.8	19.6	23.0	18.9	21.9	18.3	21.1	4.5	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.3	21.2	17.7	12.7	20.4	17.1	12.2	19.9	56.0	22.7	53.7	22.2	51.8	21.2	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.4	6.7	DB	3.5	36.7	0.7	3.2	3.0	39.0	2.6	40.9	2.2	42.7	1.6	45.0	
			WB	2.6	21.4	0.9	1.7	2.0	22.7	1.4	23.7	0.9	24.7	0.3	25.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	17.6	17.4	17.0	16.1	14.5	13.6	13.6	13.0	13.9	15.2	16.0	17.3
	DBStd	2.88	1.76	1.97	1.84	3.36	2.57	2.85	3.70	2.48	1.86	2.00	1.47	1.59
	HDD10.0	4	0	0	0	0	0	1	3	1	0	0	0	0
	HDD18.3	1160	36	36	51	86	125	149	158	168	137	103	71	41
	CDD10.0	1984	234	207	216	184	141	107	115	94	116	161	182	227
	CDD18.3	98	12	10	8	19	7	6	12	4	3	5	2	9
	CDH23.3	524	8	9	31	154	41	56	83	37	44	51	8	2
(13)	CDH26.7	176	0	1	12	63	10	14	22	8	19	24	3	0
(14)	Wind	WSAvg	3.0	3.0	3.0	2.8	2.8	2.5	2.7	2.8	3.0	3.3	3.4	2.9
(15) Precipitation	PrecAvg	68	2	2	4	9	7	13	10	9	4	5	2	3
	PrecMax	120	26	10	19	46	45	47	43	30	21	27	16	32
	PrecMin	20	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	29	5	3	5	13	9	12	9	9	5	6	4	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.4	25.2	26.7	33.5	28.2	28.6	29.2	27.6	30.2	31.0	23.0	23.6
		MCWB	20.3	19.8	19.3	18.3	14.9	14.7	14.9	12.8	15.2	14.6	16.5	19.3
	2%	DB	22.7	23.0	22.3	27.5	23.3	24.4	25.7	23.2	21.8	22.6	20.7	22.1
		MCWB	19.0	19.3	17.9	16.5	13.4	12.9	13.4	11.9	12.6	14.0	16.5	18.4
	5%	DB	21.5	21.7	20.8	23.7	20.3	20.8	22.6	19.7	17.9	19.0	19.7	21.2
		MCWB	18.2	18.3	17.4	15.7	13.5	11.5	12.3	11.4	13.6	15.1	16.1	17.8
	10%	DB	20.6	20.6	19.8	19.9	18.0	17.6	19.8	16.8	16.7	18.0	18.8	20.3
		MCWB	17.7	17.6	16.8	15.3	14.2	12.5	11.4	12.4	13.2	14.7	15.6	17.2
	0.4%	WB	20.6	20.5	19.7	19.1	17.3	15.9	15.7	14.7	15.6	17.7	18.0	19.6
		MCDB	23.6	23.7	24.8	30.2	21.1	24.9	25.4	19.0	24.5	24.6	20.6	22.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.6	19.4	18.6	17.7	15.8	14.7	14.5	13.9	14.6	16.2	17.2	18.7
		MCDB	22.0	22.2	21.5	24.3	18.9	19.2	21.2	17.6	18.5	19.6	19.8	21.5
	5%	WB	18.8	18.6	17.8	16.7	15.1	14.1	13.7	13.4	14.0	15.5	16.6	18.2
		MCDB	21.0	21.1	20.2	21.0	17.9	17.2	18.5	16.6	17.1	18.4	19.1	20.6
	10%	WB	18.0	18.0	17.1	16.0	14.6	13.6	13.0	12.9	13.5	14.9	16.0	17.6
		MCDB	20.3	20.1	19.3	19.2	17.1	16.2	16.8	15.7	16.2	17.6	18.3	19.9
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.2	5.8	7.0	6.8	7.9	8.2	7.8	7.0	6.4	5.9	5.8
		MCDBR	6.6	6.9	7.0	13.3	11.1	13.3	13.5	13.9	10.0	8.2	7.0	6.3
		MCWBR	3.8	3.9	3.9	5.2	5.3	5.7	5.5	6.4	5.6	4.4	4.0	3.7
	5% WB	MCDBR	6.0	6.3	6.4	8.7	7.5	8.3	9.1	8.3	7.7	7.0	6.1	5.9
		MCWBR	3.6	3.8	3.8	4.3	4.9	5.4	5.4	5.6	4.9	4.1	3.8	3.5
(40) Clear-Sky Solar Irradiance	taub		0.353	0.361	0.348	0.326	0.307	0.298	0.306	0.326	0.346	0.355	0.338	0.342
	taud		2.488	2.486	2.510	2.564	2.582	2.568	2.535	2.473	2.388	2.394	2.493	2.496
	Ebn at Noon		988	962	943	916	889	874	879	901	928	956	996	1001
	Edh at Noon		116	113	104	89	79	76	81	96	116	123	115	116
(44) All-Sky Solar Radiation	RadAvg		8.20	7.47	6.37	4.96	3.85	3.39	3.67	4.48	5.73	7.03	8.08	8.54
	RadStd		0.35	0.32	0.20	0.21	0.15	0.14	0.16	0.18	0.23	0.22	0.26	0.34

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page